Continual Learning for Hate Speech Detection

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Introduction

Explicit Hate Speech

BILD

Based on Zheng et al. (2024)

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HATE SPEECH DETECTION

Transfer Learnig for HS Detection

*What Did You Learn To Hate? A Topic-Oriented Analysis of Generalization in Hate Speech Detection (**Bourgeade et al. 2023**)*

- Study generalization across different popular datasets dealing with topic-specific/topic-generic HS in English tweets.
- **Cross topic generalization:** train on one dataset, test on all the datasets.
- Limitations: **noisy labels** and topic-specific datasets are only annotated for that type of hate, ie: you could find a instance of racism annotated as not hateful within a misogyny specific dataset.

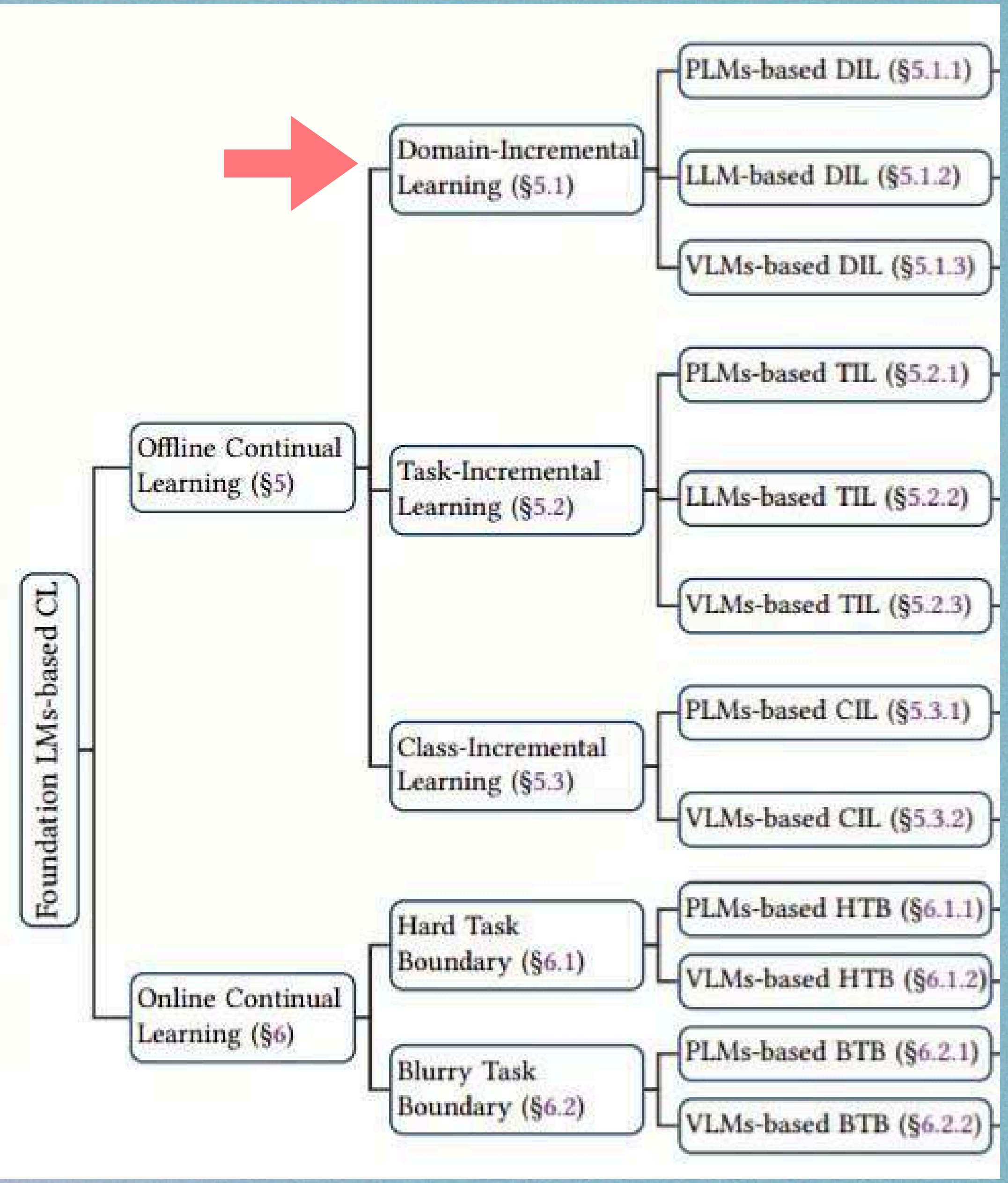
Why Continual Learning?

- **Continual Domain Adaptation** from different types of HS to one another as data becomes available.
- From a linguistics POV: **diachronic language evolution** and lexical change = shift in embeddings' distributions.
- **Emerging types of hate** across time influenced by socio-political changes.

2.
CONTINUAL LEARNING

Continual Learning?

- Training paradigm to learn new tasks **incrementally** without accessing previous data while avoiding **catastrophic forgetting** (Yang et al. 2024; Ke and Liu, 2023).
- **Stability - Plasticity** Dilemma.
- Our CL scenario: **Offline Domain-Incremental** Binary Classification of HATEFUL/NOT HATEFUL tweets.



LM-based CL Taxonomy (Yang et al. 2024)

CL Techniques

- Different types: regularization, architecture, replay, distillation....
- Implemented:
  - Regularization-based: Elastic Weight Consolidation (**EWC**) (Kirkpatrick et al. 2017), Memory Aware Synapses (**MAS**) (Aljundi et al. 2018).
  - Distillation-based: Learning without Forgetting (**LwF**) (Li and Holem 2017).
  - Replay-based: Average Gradient Episodic Memory (**A-GEM**) (Chaudhry et al. 2019).
- Our best performer: **A-GEM.**

EWC and MAS

- Regularization techniques: add a penalization to the current loss based on the parameter's importance (computed at the end of each task).
- EWC → importance through the Fisher Information Matrix (how much the loss changes wrt changes in the parameter).
- MAS → importance through the contribution of each parameter to the predictions (how much the prediction changes wrt changes in the parameter).
- Both apply a lambda hyperparam to scale the penalty.

LwF

- Distillation approach.
- Saves a copy the model after training on the previous task.
- Compute a distillation loss based on the current task batch by calculating the KL divergence between the pseudo-probabilities of teacher (old model) and the student (model during current task).
- It applies a temperature hyperparam to the softmax and a lambda to balance the loss at current time and the distillation loss.

A-GEM

- Replay / rehearsal method.
- Sample data into a memory buffer.
- **Replay the memory buffer** at current time to get reference gradients for old tasks.
- If the product of the current and reference gradients is **negative or zero**, project the old gradients onto the new ones.

procedure TRAIN($f_\theta, \mathcal{D}^{train}, \mathcal{D}^{test}$)
    $\mathcal{M} \leftarrow \{\}$
    $A \leftarrow 0 \in \mathbb{R}^{T \times T}$
    for $t = \{1, \cdots, T\}$ **do**
        for $(\mathbf{x}, y) \in \mathcal{D}_t^{train}$ **do**
            $(\mathbf{x}_{ref}, y_{ref}) \sim \mathcal{M}$
            $g_{ref} \leftarrow \nabla_\theta \ell(f_\theta(\mathbf{x}_{ref}, t), y_{ref})$
            $g \leftarrow \nabla_\theta \ell(f_\theta(\mathbf{x}, t), y)$
            if $g^\top g_{ref} \geq 0$ **then**
                $\tilde{g} \leftarrow g$
            else
                $\tilde{g} \leftarrow g - \frac{g^\top g_{ref}}{g_{ref}^\top g_{ref}} g_{ref}$
            end if
            $\theta \leftarrow \theta - \alpha \tilde{g}$
        end for
        $\mathcal{M} \leftarrow$ UPDATEEPSMEM($\mathcal{M}, \mathcal{D}_t^{train}, T$)
        $A_{t,:} \leftarrow$ EVAL($f_\theta, \mathcal{D}^{test}$)
    end for
    return f_θ, A
end procedure

A-GEM training algorithm (Chaudhry et al., 2019)

CL Evaluation

- Each task is evaluated after every task in the stream has completed training.
- **Performance matrix** of **n_tasks x n_tasks** dimensions.

	Training Stream					
DS Testing	**DS 1**	**DS 2**	**DS 3**	**DS 4**	**DS 5**	**DS 6**
DS 1	**In Training**	Training Passed	Training Passed	Training Passed	Training Passed	Training Passed
DS 2	Zero Shot	**In Training**	Training Passed	Training Passed	Training Passed	Training Passed
DS 3	Zero Shot	Zero Shot	**In Training**	Training Passed	Training Passed	Training Passed
DS 4	Zero Shot	Zero Shot	Zero Shot	**In Training**	Training Passed	Training Passed
DS 5	Zero Shot	Zero Shot	Zero Shot	Zero Shot	**In Training**	Training Passed
DS 6	Zero Shot	Zero Shot	Zero Shot	Zero Shot	Zero Shot	**In Training**

Performance Matrix Example

CL Evaluation

- Column: **performance of each dataset** after being trained on current dataset.
- Row: **evolution** of the performance in a dataset across the stream.

	Training Stream					
DS Testing	**DS 1**	**DS 2**	**DS 3**	**DS 4**	**DS 5**	**DS 6**
DS 1	**In Training**	Training Passed	Training Passed	Training Passed	Training Passed	Training Passed
DS 2	Zero Shot	**In Training**	Training Passed	Training Passed	Training Passed	Training Passed
DS 3	Zero Shot	Zero Shot	**In Training**	Training Passed	Training Passed	Training Passed
DS 4	Zero Shot	Zero Shot	Zero Shot	**In Training**	Training Passed	Training Passed
DS 5	Zero Shot	Zero Shot	Zero Shot	Zero Shot	**In Training**	Training Passed
DS 6	Zero Shot	Zero Shot	Zero Shot	Zero Shot	Zero Shot	**In Training**

Performance Matrix Example

CL Metrics (Yang et al. 2024)

- **LAST**: average performance after finishing training all tasks.
- **Forward Transfer** (FWT): how much training on previous tasks improves zero shot performance on future tasks.
- **Backward Transfer** (BWT): measures forgetting of already trained tasks after training new tasks.

$$Last = \frac{1}{N}\sum_{i=1}^{N} R_{N,i}$$

$$FWT = \frac{1}{N-1}\sum_{i=2}^{N} R_{i-1,i} - R_{0,i}$$

$$BWT = \frac{1}{N-1}\sum_{i=1}^{N-1} R_{N,i} - R_{i,i}$$

CL HSD State of the Art

- ***Qian et al. (2021):*** CL pioneer for hate speech, given a tweet -> rank hate speech groups that may have produced it.
- ***Omrani et al. (2024)***: benchmark for problematic content and tested with BART + adapters and EWC.
- ***Lin et al. (2025):*** debiasing framework with experience replay.
- ***Kang et al. (2024)***: toxicity detection for perturbated text, only paper using LLMs.

CL HSD State of the Art

Lack of Continual Learning *culture*:

- Don't report standard evaluation metrics.
- No discussion on how tasks affect each other.
- Little variety in terms of sequences and orders.

Contributions

- Study Domain Transfer in a Continual Learning setting.
- Implement 4 CL baselines: EWC, MAS, LwF and A-GEM in hate-adjacent and general PLMs for sequences of varying length.
- Study the effect of CL in Zero Shot tasks and Hate class predictions.

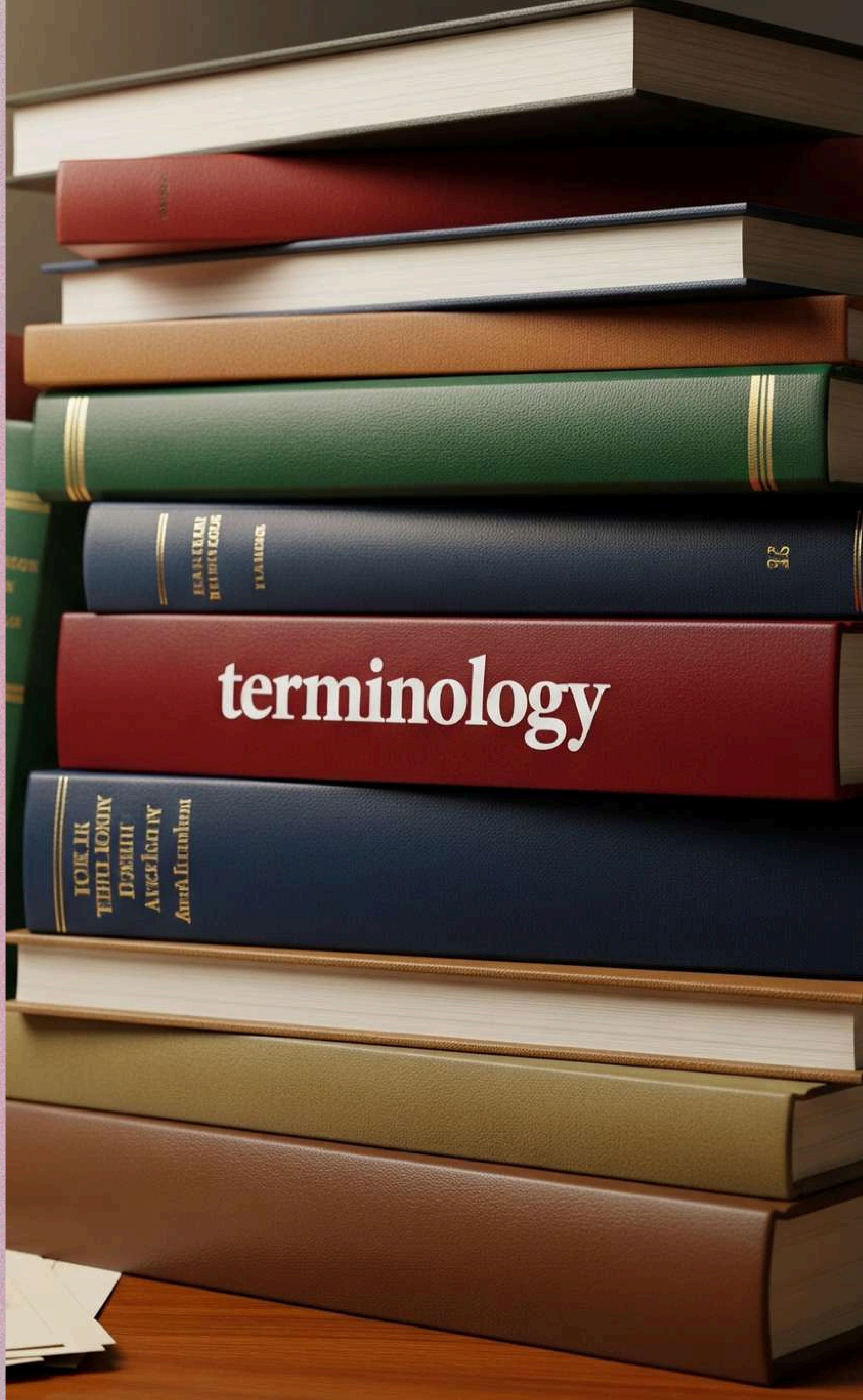
Questions

- Does CL improve domain transfer wrt sequential vanilla finetuning?
- Do certain domains transfer better than others?
- Does CL improve zero shot performance?
- What about Hate class predictions?

4. EXPERIMENTS AND RESUTLS

*Terminology

- **Dataset ~ task**
- **Datastream / stream**: the **sequence** of tasks / datasets the model will train on.
- **Performance across training**: after training each task, the model is evaluated on each of the stream's datasets' **test split**.



Experimental setup

- **3** Data streams.
- PLMs: **RoBERTa** (base), **fBERT**, **HateBERT**.
- Metrics are reported in terms of macro-F1
- **EWC, MAS, LwF and A-GEM.**
- Baselines: **SEQ-FT and M-FT.**

Model	Training Method	Data	Seq.	CL Method	CL Type	Experiment
PLM	M-FT	Union of the Data Stream	No	No	NA	Baseline
	SEQ-FT	Data Stream	Yes	No	NA	CL Baseline
	EWC	Data Stream	Yes	Yes	Regularization	CL Exp.
	MAS	Data Stream	Yes	Yes	Regularization	CL Exp.
	LwF	Data Stream	Yes	Yes	Distillation	CL Exp.
	A-GEM	Data Stream	Yes	Yes	Replay and Gradient Projection	CL Exp.

Type of PLM Models for each Experiment

Training Settings

Training Method	Epochs	Batch Size	Learning Rate	Initial CL Hyperparmeters
M-FT	8	32	1e-5	N/A
SEQ-FT	8 x number of tasks	32	1e-5	N/A
EWC	8 x number of tasks	32	1e-5	EWC LAMBDA=1500
MAS	8 x number of tasks	32	1e-5	MAS LAMBDA=1000
LwF	8 x number of tasks	32	1e-5	LwF LAMBDA=1, Temperature=2
A-GEM	8 x number of tasks	32	1e-5	Memory Size Proportion=2.5%

PLM Models Training Settings

Data

	Train/Val/Test Samples	Balance Positive Class in %	Hate Type
Davidson	4293/179/1118	26/26/26	General
Founta	44048/1836/11471	7/7/7	General
HatEval_immigrant	4500/500/1499	40/39/42	Xenophobia
Waseem_racism	10192/425/2655	15/15/15	Racism
HatEval_women	4500/500/1472	44/47/42	Misogyny
IberEval	3817/160/726	47/46/39	Misogyny
Evalita	4799/201/1000	45/45/46	Misogyny
Waseem_sexism	11159/465/2907	22/22/22	Sexism

HATE SPEECH TYPE

- General
- Racism
- Xenophobia
- Sexism
- Misogyny

Data Streams

	t1	t2	t3	t4	t5	t6
1	HatEval immigrant	Waseem racism	IberEval			
2	Evalita	Waseem racism	IberEval	HatEval immigrant		
3	Davidson	Founta	IberEval	HatEval immigrant	HatEval women	Waseem racism

Data Streams

	t1	t2	t3	t4	t5	t6
1	HatEval immigrant	Waseem racism	IberEval			
2	Evalita	Waseem racism	IberEval	HatEval immigrant		
3	Davidson	Founta	IberEval	HatEval immigrant	HatEval women	Waseem racism

➡

1. Rac. → Mis.
2. Alt. Mis. & Rac.
3. Gen. → Alt. Mis. & Rac.

BROAD RESULTS

Results by PLM and CL Technique

PLM	CL Technique	CL Metrics LAST	BWT	FWT
RoBERTa	A-GEM	**0.74** $\pm$ 0.02	0.01 $\pm$ 0.05	**0.33** $\pm$ 0.05
	EWC	0.66 $\pm$ 0.07	-0.10 ± 0.06	0.26 $\pm$ 0.08
	LwF	0.70 $\pm$ 0.06	-0.01 ± 0.07	0.31 $\pm$ 0.05
	MAS	0.67 $\pm$ 0.03	-0.01 ± 0.01	**0.33** $\pm$ 0.05
	SEQ-FT	0.68 $\pm$ 0.07	-0.06 ± 0.08	0.27 $\pm$ 0.07
HateBERT	A-GEM	0.72 $\pm$ 0.02	**0.02** $\pm$ 0.06	0.12 $\pm$ 0.04
	EWC	0.68 $\pm$ 0.04	-0.03 ± 0.05	0.09 $\pm$ 0.04
	LwF	0.68 $\pm$ 0.06	0 $\pm$ 0.06	0.14 $\pm$ 0.02
	MAS	0.63 $\pm$ 0.03	**0.02** $\pm$ 0.03	0.20 $\pm$ 0.05
	SEQ-FT	0.69 $\pm$ 0.06	-0.01 ± 0.09	0.06 $\pm$ 0.02
fBERT	A-GEM	0.72 $\pm$ 0.01	0 $\pm$ 0.02	-0.07 ± 0.11
	EWC	0.68 $\pm$ 0.03	-0.01 ± 0.03	-0.11 ± 0.08
	LwF	0.69 $\pm$ 0.04	0.01 $\pm$ 0.04	-0.10 ± 0.08
	MAS	0.60 $\pm$ 0.01	0.01 $\pm$ 0.01	-0.07 ± 0.07
	SEQ-FT	0.70 $\pm$ 0.02	0 $\pm$ 0.02	-0.11 ± 0.08

Continual Learning Results Aggregated over Experiments

- Average results for each PLM x CL technique.
- **RoBERTa A-GEM** outperforms in terms of LAST and FWT.
- **RoBERTa's transfers forward** at a much higher rate than the other PLMs.
- No model achieves a significant BWT.
- Overall, the SEQ-FT baseline already performs well.

RESULTS PER EXPERIMENT FOR **RoBERTa A-GEM**

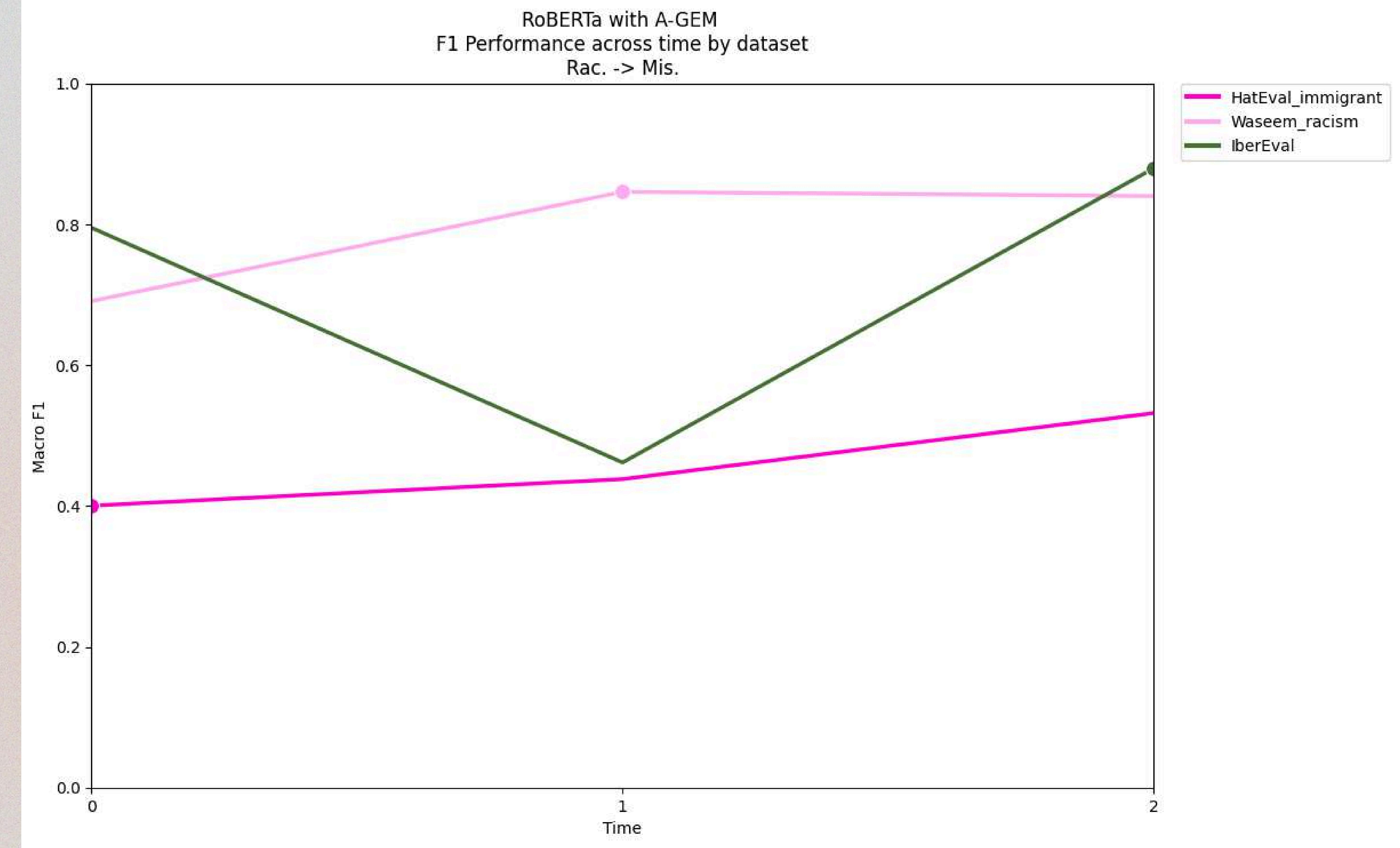
Rac. → Mis. Results

- Hardest experiment for the model.
- **Close F1 scores** wrt both baselines.
- **Better average Hate Class prediction than the SEQ-FT in the datasets part of the stream.**
- No zero-shot improvement wrt to SEQ-FT.

			M-FT		SEQ-FT		A-GEM	
Dataset	Testing Dataset	Hate Domain	F1	Hate F1	F1	Hate F1	F1	Hate F1
Trained	$\text{HatEval}_{immigrant}$	Xenophobia	0.49	**0.62**	**0.55**	0.60	0.53	**0.62**
	Waseem_{racism}	Racism	**0.85**	**0.74**	0.79	0.64	0.84	0.73
	IberEval	Misogyny	0.88	0.85	**0.89**	**0.87**	0.88	0.85
Zero Shot	Evalita	Misogyny	**0.62**	**0.60**	0.61	**0.60**	0.60	0.56
	Davidson	General	0.75	0.60	**0.79**	**0.67**	0.77	0.63
	Founta	General	0.64	0.31	**0.65**	**0.34**	0.63	0.30
	HatEval_{women}	Misogyny	0.49	0.63	0.50	0.63	**0.57**	**0.65**
	Waseem_{sexism}	Sexism	0.50	0.14	**0.65**	**0.42**	0.49	0.12

RoBERTa A-GEM LAST Performance in Rac. -> Mis.

Rac. → Mis. Performance Evolution



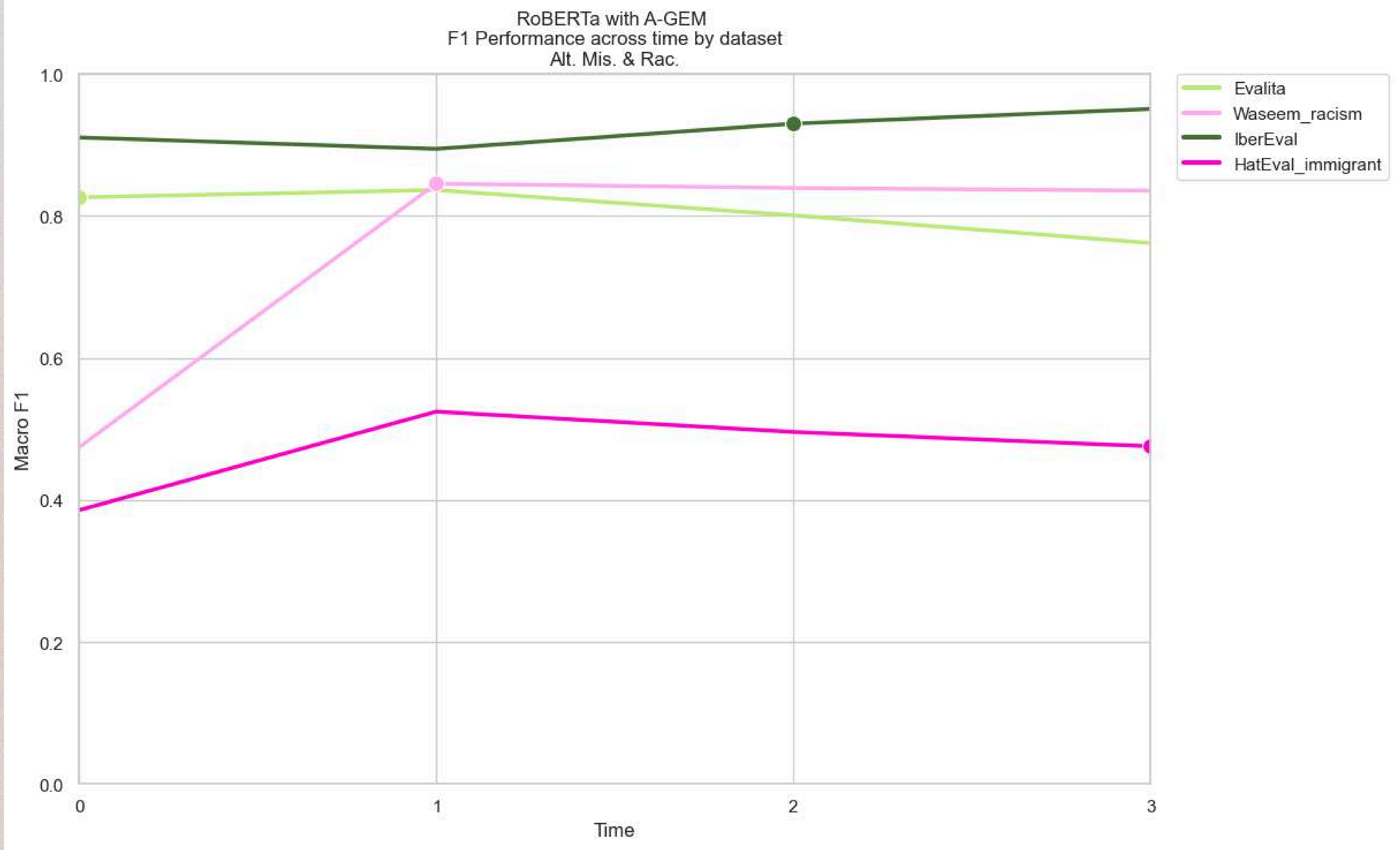
Alt. Mis. & Rac. Results

- **Better F1 and Hate F1 scores** in every dataset other than HatEval_immigrant wrt SEQ-FT.
- Better F1 zero shot in HatEval but significantly worse F1 in the general datasets.
- **Even though the stream was longer, worse performance in general datasets wrt Robert A-GEM in the first experiment.**

			M-FT		SEQ-FT		A-GEM	
Dataset	Testing Dataset	Hate Domain	F1	Hate F1	F1	Hate F1	F1	Hate F1
Trained	Evalita	Misogyny	0.70	0.69	0.63	0.71	**0.76**	**0.76**
	Waseem_{racism}	Racism	**0.84**	**0.73**	0.80	0.67	**0.84**	**0.73**
	IberEval	Misogyny	0.88	0.84	0.92	0.91	**0.95**	**0.94**
	$\text{HatEval}_{immigrant}$	Xenophobia	**0.52**	**0.62**	0.44	**0.62**	0.48	**0.62**
Zero Shot	Davidson	General	0.73	0.56	**0.83**	**0.75**	0.65	0.43
	Founta	General	0.64	0.32	**0.70**	**0.45**	0.62	0.28
	HatEval_{women}	Misogyny	0.51	0.64	0.37	0.61	**0.56**	**0.65**
	Waseem_{sexism}	Sexism	0.50	0.13	**0.68**	**0.49**	0.60	0.34

RoBERTa A-GEM LAST Performance in Alt. Mis. & Rac.

Alt. Mis. & Rac. Performance Evolution



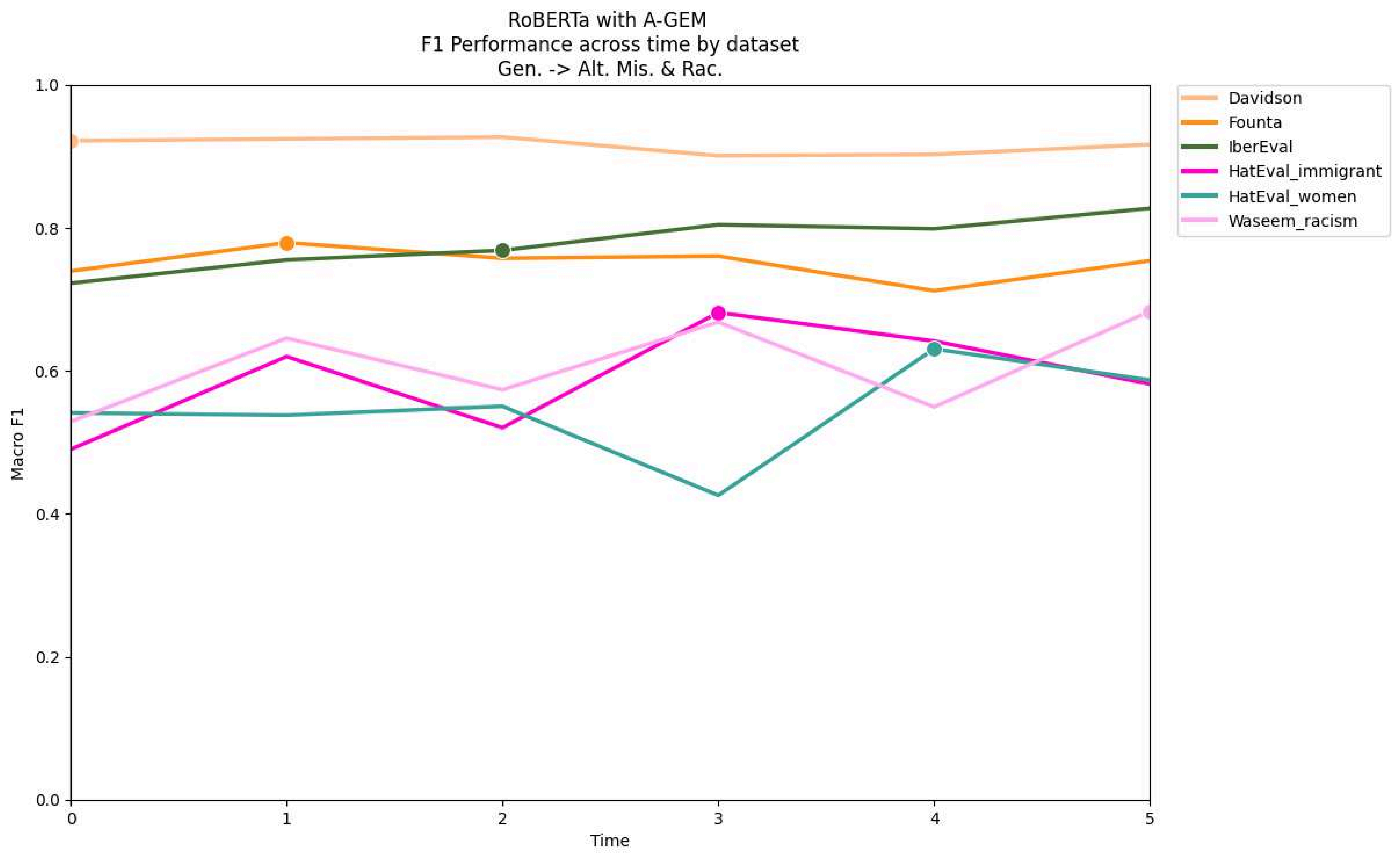
Gen. → Alt. Results

- M-FT generally outperforms the sequential models.
- A-GEM outperforms SEQ-FT and its **Hate F1** scores are much better.
- A-GEM zero shot performance also outperforms SEQ-FT's.

			M-FT		SEQ-FT		A-GEM	
Dataset Position	Testing Dataset	Hate Domain	F1	Hate F1	F1	Hate F1	F1	Hate F1
Trained	Davidson	General	0.91	**0.87**	0.75	0.59	**0.92**	**0.87**
	Founta	General	**0.77**	**0.56**	0.56	0.16	0.75	0.54
	IberEval	Misogyny	**0.86**	**0.82**	0.77	0.67	0.83	0.77
	$\text{HatEval}_{immigrant}$	Xenophobia	**0.58**	**0.64**	0.42	0.11	**0.58**	**0.64**
	HatEval_{women}	Misogyny	0.51	0.64	**0.65**	**0.66**	0.59	0.65
	Waseem_{racism}	Racism	**0.82**	**0.69**	0.51	0.09	0.68	0.44
Zero Shot	Evalita	Misogyny	**0.62**	**0.63**	**0.62**	0.50	**0.62**	0.62
	Waseem_{sexism}	Sexism	**0.53**	**0.20**	0.46	0.04	0.51	0.16

RoBERTa A-GEM LAST Performance in Gen. -> Alt. Mis. & Rac.

Gen. → Alt. Performance Evolution



5.
CONCLUSION AND FUTURE WORK

CONCLUSION

We have studied the effect of 4 CL techniques applied to 3 models in 3 setting:

- RoBERTa outperformed the other PLMs, A-GEM was the most successful technique and the effect of our best model was more noticeable in the longest sequence.
- **A-GEM**, and as we have seen, RoBERTa A-GEM**, consistently improves the prediction of the hateful messages** but Continual Learning techniques do not generally help with zero shot performance.

We are targetting **LREC 2026** for publication.

FUTURE WORK

- LLM-based Continual Learning for Implicit Targeted Hate Speech Detection
- Time-agnostic Hate Speech Detection

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CODA

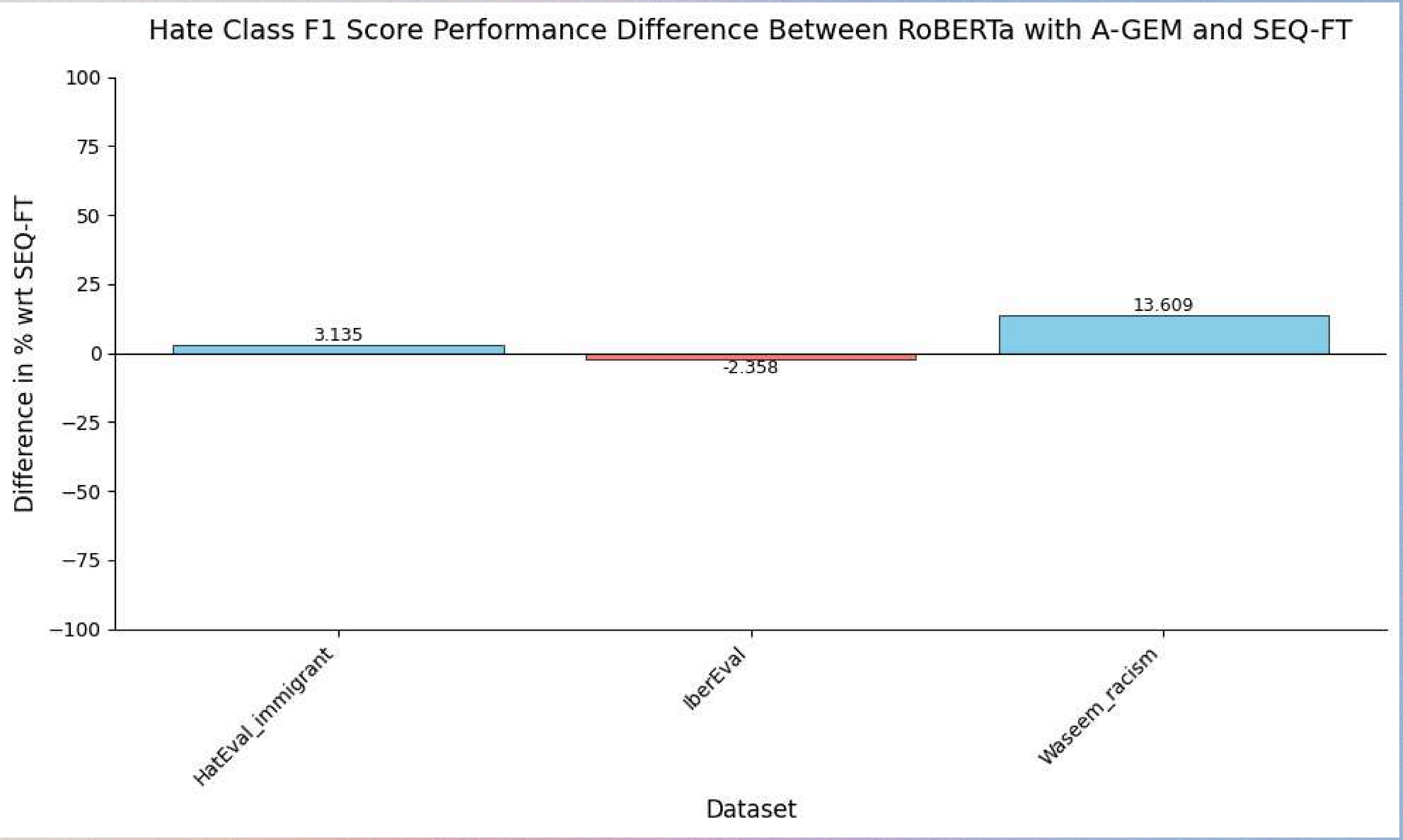
Results by CL Technique

	CL Metrics		
CL Technique	LAST	BWT	FWT
A-GEM	$\mathbf{0.73} \pm 0.02$	$\mathbf{0.01} \pm 0.04$	0.13 ± 0.18
EWC	0.67 ± 0.05	-0.05 ± 0.06	0.08 ± 0.17
LwF	0.69 ± 0.05	0 ± 0.05	0.12 ± 0.19
MAS	0.63 ± 0.04	0 ± 0.02	$\mathbf{0.15} \pm 0.19$
SEQ-FT	0.69 ± 0.05	-0.02 ± 0.07	0.07 ± 0.17

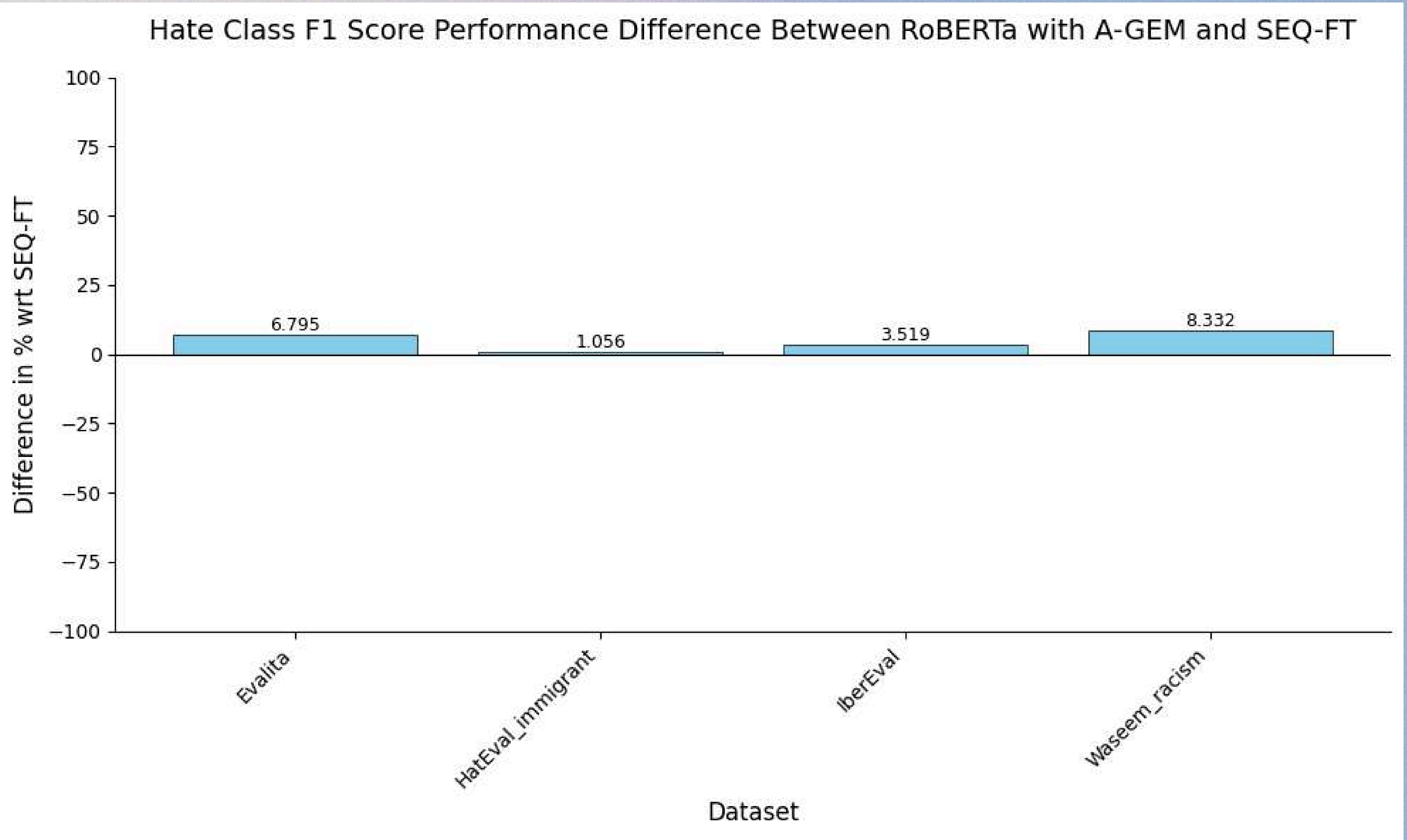
Aggregated Results by Continual Learning Technique

- Average results for each CL technique.
- **A-GEM** outperforms the other techniques.
- **A-GEM is the only technique that achieves an average positive BWT.**
- MAS excels in FWT.
- Overall, the SEQ-FT baseline already performs well.

Rac. → Mis. Hate Class Comparison in %



Alt. Mis. & Rac. Hate Class Comparison in %



Gen. → Alt. Hate Class Comparison in %